COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010	**SECTION:**	CPE21S4
DATE PERFORMED:	02/09/2026	**DATE SUBMITTED:**	02/09/2026
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ACTIVITY NO. Seatwork_7.2 **Algorithm Analysis**

1. PROCEDURE OUTPUT

What procedures did you perform?

1. Finding an Element

```
for (int i = 0; i < n; i++)
{
    if (arr[i] == target)
    {
        return i;
    }
}
```

Best Time Complexity > O(1)

Worst Time Complexity > O(n)

Explanation: the best time is O(1) because it only run once and the worst time is O(n) because it's not a n loop

2. Printing All Elements

```
for (int i = 0; i < n; i++)
{
    cout << arr[i] << endl;
}
```

Best Time Complexity >O(n)

Worst Time Complexity>O(n)

Explanation: The loop runs n time no exit so the worst and best time is both O(n)

3. Nested Loops

```
for (int i = 0; i < n; i++)
{
    for (int j = 0; j < n; j++)
    {
        cout << i << " " << j << endl;
    }
}
```

Best Time Complexity >O(n^ 2)

Worst Time Complexity>O(n^2)

Explanation: the inner loop runs n time and the outer loop runs n time so best and worst time is the same asymptotic

4. Conditional Processing

```
if (n > 100)
{
    for (int i = 0; i < n; i++)
    {
        cout << arr[i];
    }
}
else
{
    cout << "Small input";
}
```

Best Time Complexity >O(1)

Worst Time Complexity>O(n)

Explanation: even n > 100 or less we just simply use the constant one and worst time is O(n) because it only will run it n time

5. Fixed Inner Loop

```
for (int i = 0; i < n; i++)
{
    for (int j = 0; j < 10; j++)
    {
        cout << i << j;
    }
}
```

Best Time Complexity >O(n)

Worst Time Complexity>O(n)

Explanation: the outer loop runs n time but the inner loop runs 10 time only

6. Two Consecutive Loops

```
for (int i = 0; i < n; i++)
{
    cout << arr[i];
}

for (int j = 0; j < n; j++)
{
    cout << arr[j] * 2;
}
```

Best Time Complexity >O(n)

Worst Time Complexity>O(n)

Explanation: best time is the first loop and run n times so the second loop also runs n times and it becomes 2n but because of the asymptotic we ignore the number and it equals to O(n)

7. Dividing by Two

```
int x = n;

while (x > 1)
{
    x = x / 2;
}
```

Best Time Complexity >O(log n)

Worst Time Complexity>O(log n)

Explanation: its both O(log n) because it divide it little by little

8. Nested Loop With Different Limits

```
for (int i = 0; i < n; i++)
{
    for (int j = 0; j < i; j++)
    {
        cout << i << j;
    }
}
```

Best Time Complexity >O(n^2)

Worst Time Complexity>O(n^2)

Explanation: its both O(n^2) because the outer loop runs n times and the inner loop runs n times so it becomes two n and combined it and got O(n^2)

9. Early Termination

```
for (int i = 0; i < n; i++)
{
    if (arr[i] == target)
    {
        cout << "Found";
        break;
    }
}
```

Best Time Complexity >O(1)

Worst Time Complexity>O(n)

Explanation: Best time is O(1) because there's a break

10. Two-Level Search

```
for (int i = 0; i < n; i++)
{
    for (int j = 0; j < n; j++)
    {
        if (arr[i] == target)
        {
            break;
        }
    }
}
```

Best Time Complexity >O(1)

Worst Time Complexity>O(n^2)

Explanation: the best time is O(1) because it is the fastest time to get it executed, as you can see there's a break and the worst time is O(n^2)
Because it will runs n time

11. Linear Loop With Logarithmic Inner Work

```
for (int i = 0; i < n; i++)
{
    int x = n;

    while (x > 1)
    {
        x = x / 2;
    }
}
```

Best Time Complexity >O(n Log n)

Worst Time Complexity>O(n Log n)

Explanation: as you can see the outer loop runs n times and the inside each, and there's a while loop and x / 2
So it makes it divide it little by little

12. Conditional Nested Loop

```
if (arr[0] == target)
{
    cout << "Found";
}
else
{
    for (int i = 0; i < n; i++)
    {
        for (int j = 0; j < n; j++)
        {
            cout << arr[i] + arr[j];
        }
    }
}
```

Best Time Complexity >O(1)

Worst Time Complexity>O(n^2)

Explanation:the element of the first one is matches the constant target and the worst time it doesnt match the target but it is a conditional scenario so its either both best or both worst

13. Finding the Maximum

```
int max = arr[0];

for (int i = 1; i < n; i++)
{
    if (arr[i] > max)
    {
        max = arr[i];
    }
}
```

Best Time Complexity >O(n)

Worst Time Complexity>O(n)

Explanation: both best time and worst time because the code it show's only runs n times and there's no exit

14. Three Nested Loops

```
for (int i = 0; i < n; i++)
{
    for (int j = 0; j < n; j++)
    {
        for (int k = 0; k < n; k++)
        {
            cout << i + j + k;
        }
    }
}
```

Best Time Complexity >O(n^3)

Worst Time Complexity>O(n^3)

Explanation: three separate loops and each of them runs n times and they are the same best time and worst time no condition or break happens and its n x n x n so we got O(n^3)

15. Comprehensive Analysis

```
for (int i = 0; i < n; i++)
{
    if (arr[i] == target)
    {
        return i;
    }

    int x = n;

    while (x > 1)
    {
        x = x / 2;
    }
}
```

Best Time Complexity >O(1)

Worst Time Complexity>O(n log n)

Explanation: best time is O(1) because it is the fastest time to execute the code while worst time is O(n log n) because it is the slowest time to run the code

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

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Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

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